# US Payment Network Integration

Engagement Proposal | FedWire · ACH · RTP

CTO on Demand Inc.

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| Prepared for: | **bunq B.V.** |
| Prepared by: | **CTO on Demand Inc.** |
| Engagement: | FedWire + ACH + RTP Integration |
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# Contents

# 1 The Opportunity

bunq is integrating directly into three US payment networks — FedWire, ACH, and RTP — during its US banking charter conditional approval window.

CTO on Demand augments bunq's engineering team with US payment network expertise and regulatory depth that cannot be built quickly from the outside. Three senior engineers work directly alongside bunq's developers. Same environment, same pace, same Slack. CTO on Demand manages Federal Reserve, TCH, and NACHA access in parallel so bunq's team stays focused on the build.

**Key Insight**

Deliverable A production-ready ISO 20022 messaging layer across all three networks, built against official specifications from day one. Direct integrations, bunq's IP, bunq's core.

## 1.1 Programme Highlights

- √ US payment network and regulatory expertise embedded directly into bunq's engineering team
- √ Engineers with hands-on experience building payment infrastructure at JPMC and American Express
- √ Single ISO 20022 architecture across FedWire, ACH, and RTP
- √ Automated routing and failover with configurable priority — bunq sets the sequence based on speed vs. cost (default: RTP → FedWire → ACH)
- √ Phase 1 built entirely offline — no sandbox dependency, no idle time
- √ Sandbox access progressed through CTO on Demand's Federal Reserve, TCH, and NACHA relationships from week one
- √ Full QA across all rails: happy path, errors, edge cases, failover
- √ BSA/AML and OFAC integration points documented and handed to bunq
- √ 3 FTE engineers embedded; time and materials, billed monthly

## 2 The Build

### 2.1 One Architecture. Three Networks.

A unified ISO 20022 messaging layer across FedWire, ACH, and RTP, integrated directly into bunq's core banking system. One architecture, three networks, intelligent routing, automated failover.

Payments route automatically based on receiving bank network capability and bunq's internally configured priority. The default sequence is RTP → FedWire → ACH, optimised for speed. bunq can reconfigure this order at any time. On failure, a webhook fires immediately and the next network initiates without manual intervention. Every routing decision is logged and auditable.

**Table 1:** Payment Network Coverage

| Network | Message Types | Role |
|---|---|---|
| RTP (The Clearing House) | pacs.008, pacs.002, camt.035, remt.001 | Primary instant rail |
| FedWire (Federal Reserve) | pacs.008, pacs.009, camt series | Interbank & liquidity |
| ACH (NACHA) | pain.001, pacs.008 mapping | Settlement backstop |

# 3 Under the Hood

## 3.1 ISO 20022 Messaging Core

**Message Schema Registry**

XML schemas across pacs, camt, pain, and remt families. Version-controlled as networks update.

**Message Builder API**

Field-level validation, data type enforcement, and cardinality checks per specification.

**Message Parser**

Bidirectional parsing of inbound ISO 20022 XML validated against network-specific business rules.

**Validation Engine**

Seven-layer validation from syntax through full business process validation.

## 3.2 Network Adapters

**RTP : The Clearing House**

Message identification per TCH specification. Duplicate checking. Local Instrument codes (CONSUMER, BUSINESS, INTERMEDIARY). Full lifecycle: pacs.008 → pacs.002 → camt.035 including Accept Without Posting flows. Full reason code handling.

**FedWire : Federal Reserve**

pacs.008 for customer transfers. pacs.009 for FI-to-FI. ABA routing validation. FRB business rules per technical documentation. FedLine connectivity. Cutoff time management, queueing, and next-day settlement logic.

**ACH : NACHA**

Bidirectional ISO 20022 to NACHA file mapping per NACHA's official Mapping Guide. SEC codes: PPD, CCD, CTX, WEB, TEL. Same-day vs. standard window management. Return and NOC handling mapped to ISO 20022 status messages.

### 3.3 Payment Orchestration Layer

**Routing Engine**

Network selection per transaction based on receiving bank capability, live availability, and bunq's configurable cost vs. speed priority. Default: RTP → FedWire → ACH.

**Automated Failover**

On failure: immediate webhook, next network initiated, full decision chain logged. Priority order fully customisable.

**Routing Database**

Receiving bank capabilities by routing number, maintained throughout.

**Webhook & Event System**

Real-time streaming across all payment lifecycle events.

**Monitoring & Observability**

Dashboards, alerting, performance metrics, reconciliation reporting.

## 4 How We Get There

**Business Value**

Investment $50,000 per month. 3 senior engineers embedded with bunq's team. One fixed monthly rate. No variable billing. No surprises. 6-month minimum commitment, 12-month expected engagement.

**Table 2:** Delivery Phases

| Phase | Timeline | Scope |
|---|---|---|
| 1: Core & Adapters | Months 1–3 | ISO 20022 core built against official specification. FedWire, ACH, and RTP adapters. Core banking integration layer jointly designed with bunq's team. |
| 2: Orchestration & Sandbox | Months 3–6 | Routing engine. Failover logic. Webhook system. Sandbox integration and end-to-end testing across all three networks. |
| 3: QA & Hardening | Months 6–9 | Full regression. Security review. Performance testing. Compliance documentation handed to bunq. |
| 4: Production & Hypercare | Months 9–12 | Network certification. Go-live support. Post-launch stabilisation. |

Phase 1 has zero external dependencies. All adapters are built against official documentation before sandbox credentials are needed. Sandbox access requests are initiated through CTO on Demand's network relationships from week one. If delayed, Phase 2 shifts. Total engagement unchanged.

**Table 3:** Engagement Pricing

| Term | Amount |
|---|---:|
| Monthly rate | $50,000 |
| Minimum commitment (6 months) | $300,000 |
| Full engagement (12 months) | $600,000 |

## 5 Who You Get

### 5.1 Morgan Reed

*Technical Lead & Engagement Manager*

Former CTO, Foundations Investment Advisors. Led technology and compliance infrastructure across two financial firms managing over $5B AUM. Western Alliance Bank: engineering solutions delivered directly into regulated US banking infrastructure. USAA: vendor API integrations within Article 7 compliance frameworks. American Express: payment application development and the Plenti mobile payments platform. Responsible for architecture decisions, delivery coordination, and direct engagement with the Federal Reserve, TCH, and NACHA throughout.

### 5.2 Royce Carbowitz

*Senior Payment Engineer*

Former VP and Lead Software Engineer, J.P. Morgan Chase. American Express. Built payment APIs processing millions of daily transactions across Visa, Amex, and Mastercard networks. Led infrastructure modernisation at JPMC to 99.9% uptime. Spring Boot, Kafka, Kubernetes, AWS. U.S. Navy, Nuclear Power trained.

### 5.3 Integration Engineer

*+1 Dedicated Resource*

Core banking integration layer, orchestration engine, API layer, and webhook system.

### 5.4 Compliance Practice

Licensed compliance officers with deep experience in US federal and state banking frameworks. Active relationships with the OCC, Federal Reserve, FinCEN, TCH, and NACHA. Every compliance deliverable is owned by a named compliance professional.

## 6 Testing

Testing is built into every phase of the engagement, not bolted on at the end.

**Unit Testing**
Every piece of code goes through unit testing before anything else. Non-negotiable.

**Automated Testing**
An automated test suite runs across all payment flows — standard transactions, failure scenarios, edge cases, and failover sequences across RTP, FedWire, and ACH.

**Manual Testing**
A dedicated manual tester reviews everything the automated suite may not fully cover. Every gap identified is documented and resolved before moving forward.

**Performance & Load Testing**
The system is tested under production-grade transaction volumes. Throughput, latency, and reliability validated before any production connectivity is established.

**Security Review**
Encryption, key management, and audit trail validation reviewed as part of Phase 3. BSA/AML and OFAC integration points tested, documented, and handed to bunq.

**Penetration Testing**
CTO on Demand recommends engagement with an independent third-party pen tester for objectivity. Recommendations for certified providers available on request.

**Certifications**
SOC2 and PCI certifications conducted by an independent certified body. CTO on Demand builds to those specifications throughout and prepares the full documentation package.

## 7 Compliance & Regulatory

The technical integration and the regulatory framework are not separate workstreams. Every payment rail built in this engagement has compliance obligations attached to it. CTO on Demand carries both.

### 7.1 Included: Compliance Documentation Audit

Before anything is built, CTO on Demand conducts a full review of bunq's existing compliance documentation — every policy, procedure, framework, and control assessed against US regulatory standards (BSA/AML, FinCEN, OFAC, and network-specific requirements). Included at no additional charge.

Deliverables:

- Complete gap analysis across all existing documentation
- Specific recommendations for what needs to be created, updated, or restructured
- Prioritised action plan aligned to the conditional approval timeline
- Clear picture of what bunq's CCO needs in place before the regulator's on-site examination

### 7.2 One-Off Compliance Documents

Each document built from CTO on Demand's library of 50+ US-specific templates, customised to bunq's business model, digital onboarding workflows, and cross-border payment structure. Fixed fee, scoped on request.

- BSA/AML Programme — built to pass examination, not just satisfy a checklist
- KYC/KYB Procedures — tailored to bunq's digital onboarding model
- Sanctions Screening Controls — OFAC compliance framework
- Transaction Monitoring Framework — rules-based and AI-assisted, aligned to FinCEN guidance
- Consumer Complaint Policy — US-specific handling procedures
- Recordkeeping Procedures — BSA, FinCEN, and state-level obligations
- Regulatory Reporting Framework — CTR, SAR, and state-specific obligations
- Payment Rail Strategic Advisory — RTP vs. FedNow analysis, sequencing strategy

### 7.3 Ongoing Compliance Support

CTO on Demand deploys a compliance team working alongside bunq's in-house CCO. Month-to-month, no long-term commitment.

**Table 4:** Compliance Services & Fees

| Service | Fee |
|---|---|
| Full compliance documentation audit | Included |
| One-off policy and procedure documents | Fixed fee, scoped on request |
| Payment rail strategic advisory | Fixed fee, scoped on request |
| Ongoing compliance support | $10,000/month |

Ongoing support covers policy maintenance, regulatory change management, BSA/AML programme management, examination readiness, independent AML audit coordination, and payment network compliance integration.

## 8 Why CTO on Demand

bunq has three options. This section addresses all three directly.

### 8.1 Option 1: Build In-House

Hiring US payment network engineers with experience on a custom core is exceptionally rare. Most US payment engineers have worked exclusively with known cores — FIS, Temenos, Thought Machine. Finding the right people while a conditional approval window is running is a multi-month process with no guaranteed outcome.

The deeper issue is the commitment: 3–4 senior engineers at $180,000–$250,000 base salary each. $600,000–$1,000,000 in annual headcount for a finite integration project. Once the integrations are built, what do those engineers do? A permanent headcount commitment for a time-bounded problem runs against everything bunq stands for. And an engineering hire does not come with compliance expertise — that is a separate problem on the same ticking clock.

### 8.2 Option 2: Pure Consultancy

A consultancy advises. bunq still does all the building. Their playbooks and integration experience are based on FIS, Temenos, or similar — not a custom core. There is also no sandbox access. A consultancy can tell bunq how to apply for credentials; that is not the same as having the relationships to obtain them.

### 8.3 Option 3: CTO on Demand

**Key Insight**

The Combination That Matters Technical execution and regulatory knowledge operate as a single practice. CTO on Demand has built its own custom core banking system. Direct payment network integrations on custom cores are not new territory.

- Technical — 70–80% of the work is built against official specifications before sandbox access is required. No dependency on any core vendor.
- Regulatory — Active relationships with the Federal Reserve, OCC, FinCEN, TCH, and NACHA. Sandbox access progressed from week one.
- Compliance — 50+ US-specific templates customised to bunq's model. Independent audit included at no charge.
- Knowledge transfer — CTO on Demand builds alongside bunq's team, trains bunq's engineers throughout, and hands over full documentation and runbooks. bunq owns everything. No ongoing dependency.

### 8.4 References

CTO on Demand works with larger fintechs and bank charter applicants across MTL programmes and payment network integrations. References available on request. Morgan and Royce have built these exact integrations in their previous roles at J.P. Morgan Chase, American Express, Western Alliance Bank, and USAA.

## 9 What to Expect

**Core Banking Integration Complexity**

Integration architecture jointly scoped with bunq's team in the first two weeks. Clean abstraction boundaries mean complexity on bunq's side does not impact the integration layer.

**Network Specification Changes**

Schema registry is version-controlled. Adapters built for backward compatibility. Release notes monitored throughout.

## 10 What's In, What's Out

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|---|---|
| In Scope | FedWire, ACH, RTP. USD domestic payments. BSA/AML and OFAC integration points documented and handed to bunq. |
| Out of Scope | Zelle (future phase), FedNow (future adapter), Request for Payment: pain.013 (future enhancement), cross-border and SWIFT. |

## 11 What We Need From bunq

- bunq's core banking system exposes or will expose interfaces sufficient for the payment messaging layer. Architecture jointly designed.
- bunq provides dedicated engineering counterparts for integration design and development.
- USD domestic payments only. Cross-border, FX, and multi-currency excluded.
- BSA/AML, OFAC, fraud, and KYC remain bunq's responsibility. CTO on Demand designs and documents integration points.

- ISO 20022 documentation procured by CTO on Demand and billed to bunq at actual cost, no markup.

## 12 Let's Go

1. Technical session with bunq's engineering leads. Architecture jointly scoped.
2. Commercial terms and rate card confirmed.
3. Sandbox access applications initiated alongside Phase 1.

**Ready when you are.**